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PROFESSIONAL SUMMARY

Computer Science student with applied **machine learning** experience building **production-ready ML systems** — including a GraphSAGE-based graph neural network achieving **99.57% accuracy** with inference optimized to **~120ms per graph** for real-time deployment. Experienced integrating ML and inference pipelines into full-stack applications (FastAPI, async task queues, PostgreSQL) and deploying them at scale with **Docker, AWS, and CI/CD**. Strong foundation in algorithms, data structures, and system design, backed by production software engineering experience shipping **6+ merged pull requests** as a developer intern.

EDUCATION

Canara Engineering College, Mangalore
CGPA: 8.26

B.E in Computer Science & Engineering
2026

KSEAB (SSLC/PUC)

SSLC: 88.96% — PUC: 92.96%

PROJECTS

GraphSentinel — Graph Neural Network Cyber Defense System

Python, PyTorch Geometric, GraphSAGE

- Designed, trained, and optimized a GraphSAGE-based intrusion detection model on CICIDS-2017, choosing an inductive architecture over GCN specifically to classify previously unseen IPs at inference time.
- Engineered a **7-feature behavioral signal set** (port entropy, byte asymmetry, out-degree, etc.) from raw traffic data, and resolved severe class imbalance (128K vs. 1,956 samples) using hybrid under-sampling with SMOTE and weighted cross-entropy loss.
- Achieved **99.57% accuracy** and **0.9975 AUC-ROC**; optimized inference speed to **~120ms per graph** to fit a production real-time detection budget of 5 seconds.

HVSapp — Healthcare Handoff Validation Platform

FastAPI, React Native, PostgreSQL, Celery

- Integrated an **ASR (speech-to-text) transcription model** into a production clinical application, decoupling inference into **async Celery task queues** to keep the API responsive under real-time load.
- Applied targeted database indexing and async task decoupling to **scale API responsiveness** under concurrent clinical workflows.
- Implemented PHI field encryption and medication safety tracking via Alembic migrations, structuring the schema around clear domain boundaries (auth, medication, patients, transcription).

EXPERIENCE

Orbit — Intern, Technical Team Member

Feb 2026 – Present

- Built and deployed backend services (Node.js/Express) for a production location-based social platform, engineering real-time infrastructure via WebSockets and geolocation-based discovery services.
- Deployed and maintained the backend in production on Render with Sentry error monitoring and Docker-based environment consistency; shipped mobile builds via Expo Application Services (EAS).
- Merged **5+ pull requests** into production, including resolving **API rate-limiting** issues to keep services stable under load.

ACHIEVEMENTS & LEADERSHIP

Core Unleashed Hackathon — Team Lead, “Digital Sentinels”

Apr 2025

- Ranked among the **Top 10 teams** out of **72 teams** and pitched our solution to the final judging panel after advancing through multiple evaluation rounds.
- Engineered a stateless backend with a deterministic finding-ID system and a **3-tier bounding-box** fallback (exact character → line-level → sidebar-only) to keep redaction accurate despite imperfect OCR spatial mapping.

Feature Development & Deployment

Ongoing

- Merged **5+ pull requests** into Orbit’s production codebase, shipping UI improvements, new features, and backend fixes.
- Redesigned the landing page and signup flow UI, and built a **location-picker dropdown that reverse-geocodes** a user-dropped map pin into an address.
- Implemented **event creation with feed-level visibility** and resolved **API rate-limiting** issues to keep the mobile client responsive under load.

TECHNICAL SKILLS

- Languages:** Python, Java, JavaScript, SQL
- ML & Data:** PyTorch, PyTorch Geometric (GraphSAGE), Scikit-learn, OpenCV, MediaPipe, Tesseract OCR
- Backend & Cloud:** FastAPI, Flask, Node.js, Express, AWS (EC2, S3, VPC), PostgreSQL, MongoDB, Redis
- DevOps & Tools:** Docker, Git/GitHub, CI/CD basics, Celery, WebSockets, Linux Administration
- Frontend & Mobile:** React.js, React Native, Redux, Tailwind CSS